

PGDAIML – CAPSTONE PROJECT

○ TECHNICAL PRESENTATION

AGENDA

Problem & Data Overview

EDA Key Findings

Feature Engineering Approach

Model Selection & Methodology

Feature Importance & Explainability

Limitations & Future Work

PROBLEM AND DATA OVERVIEW

INTRUSION DETECTION SYSTEMS (IDS), PROMOTING THE ADVANCEMENT OF ROBUST AND ADAPTIVE SECURITY SOLUTIONS TAILORED FOR THE DYNAMIC LANDSCAPE OF REAL-TIME IOT ENVIRONMENTS.

- Project Objective: Detect anomalies in event data by analyzing log records to distinguish normal and anomalous activities, aiming to identify potential security threats and operational concerns within a network.
- Enhance Intrusion Detection Systems (IDS) by applying anomaly detection insights, supporting proactive and real-time IoT network security solutions.
- Evaluation Metrics: Effectiveness of anomaly classification models will be measured using Precision, Recall, F1 Score, and Area Under the ROC Curve (AUC).
- Machine Learning Algorithms Considered: Random Forest, XGBoost, and Logistic Regression will be compared to find the best model for anomaly detection.
- Data Source: Uses RT-IoT2022, a proprietary dataset from a real-time IoT infrastructure, featuring diverse IoT devices and simulated network attacks for realistic traffic patterns.
- Devices in the dataset include ThingSpeak-LED, Wipro-Bulb, and MQTT-Temp, representing typical IoT deployments.
- Simulated attack scenarios in the dataset include Brute-Force SSH, DDoS (via Hping and Slowloris), and reconnaissance/exploitation with Nmap.
- Data granularity ensured using Zeek network monitoring and Flowmeter plugin, capturing detailed bidirectional traffic attributes.
- RT-IoT2022 serves as a critical resource for advancing IDS capabilities and developing adaptive security solutions for dynamic, real-time IoT environments.

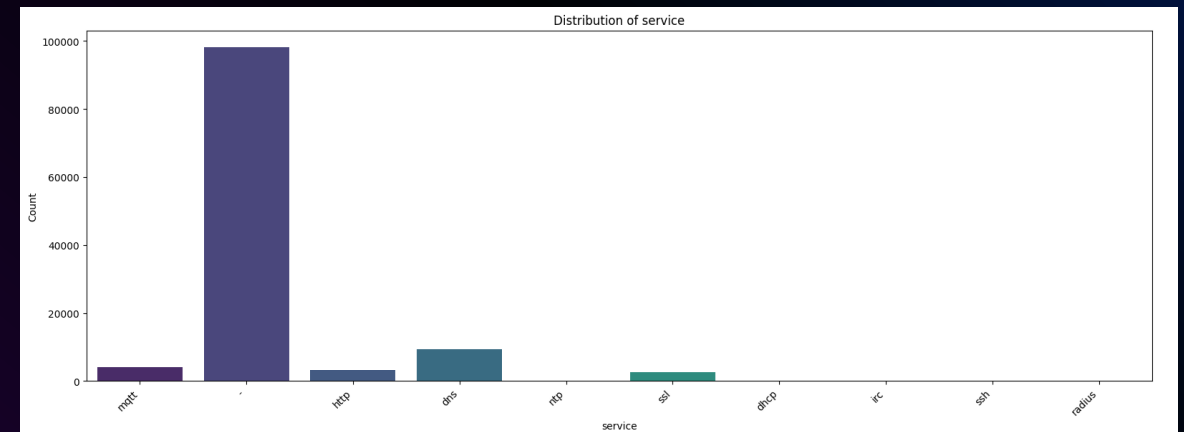
EDA KEY FINDINGS

DISTRIBUTION OF CATEGORICAL FEATURES

USING UNIVARIATE AND BIVARIATE ANALYSIS

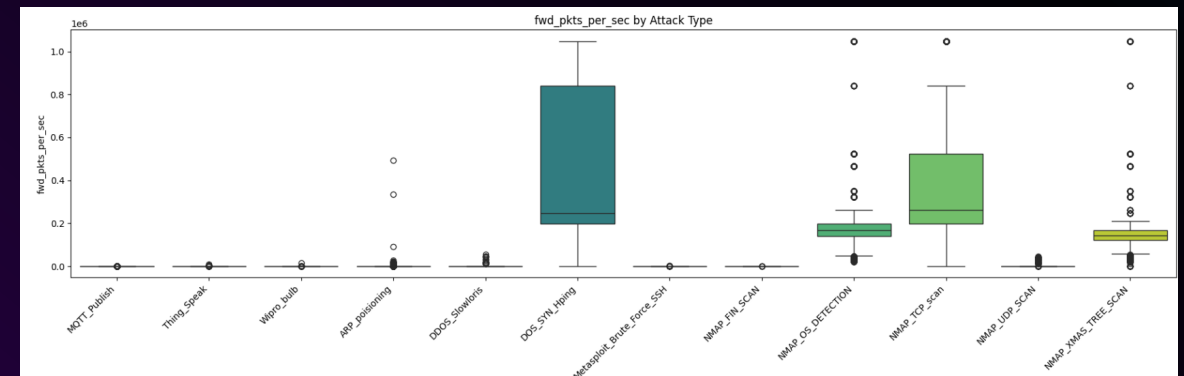
Univariate Analysis:

- **Attack_type Distribution:** The data shows marked class imbalance, with some attack types (e.g., 'DOS_SYN_Hping', 'MQTT_Publish') occurring far more frequently—an important factor for model development due to potential bias.
- **Categorical Features (proto, service):** Protocols like 'tcp' dominate network traffic, and the '-' service category is especially frequent.



Bivariate Analysis:

- **Numerical Features vs. Attack_type:** Box plots reveal that features such as flow_duration, fwd_pkts_per_sec, and payload_bytes_per_second vary noticeably across attack types, with outliers indicating anomalous behavior.
- **Categorical Features vs. Attack_type:** Certain attack types are closely linked to specific protocols and services, e.g., 'DOS_SYN_Hping' mostly uses 'tcp', while 'MQTT_Publish' heavily involves the 'mqtt' service.



FEATURE ENGINEERING APPROACH

FEATURE ENGINEERING

SCALING: ROBUSTSCALER

ENCODING: ONE-HOT ENCODING

FILTER METHOD: MUTUAL INFORMATION

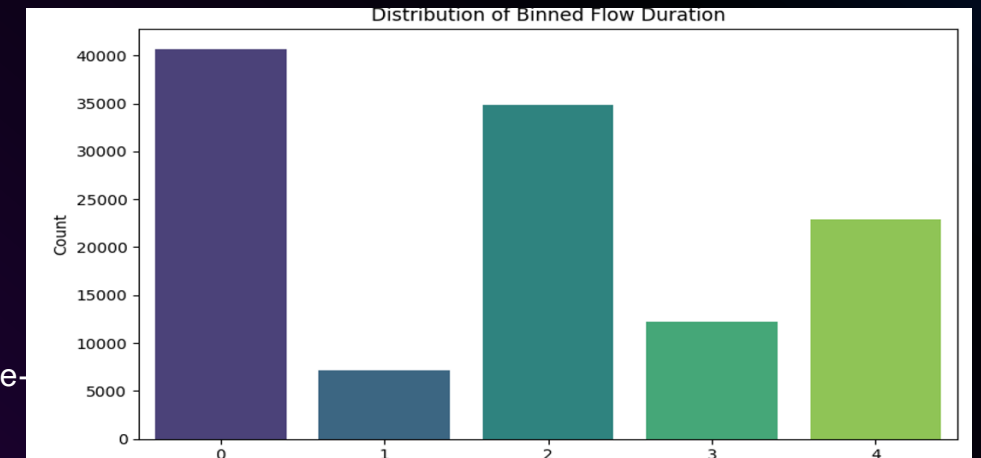
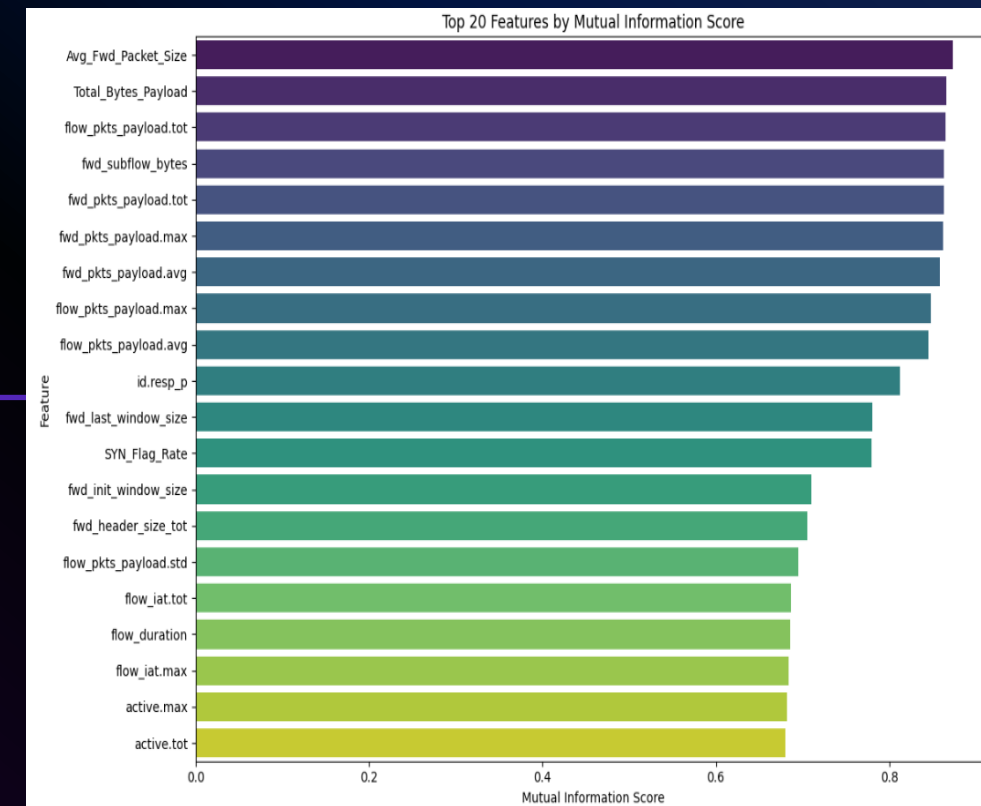
WRAPPER METHOD: RECURSIVE FEATURE ELIMINATION

Several new domain-specific features were created to enhance the model's predictive power:

- Total_Packets
- Total_Bytes_Payload
- Fwd_Bwd_Packet_Ratio
- Fwd_Bwd_Data_Packet_Ratio
- Avg_Fwd_Packet_Size
- Avg_Bwd_Packet_Size
- SYN_Flag_Rate, FIN_Flag_Rate, RST_Flag_Rate
- flow_duration_binned: flow_duration was binned into 5 equal-frequency categories.

Categorical Feature Handling

- Categorical features (proto, service, and flow_duration_binned) were transformed using One-Encoding, significantly increasing the number of features in X to 108.
- The target variable Attack_type in y was Label Encoded into 12 numerical classes (0-11).



MODEL SELECTION AND METHODOLOGY

MODEL TRAINING AND EVALUATION

- The data was split into training (80%) and testing (20%) sets with stratification. Several classification models were trained and evaluated on the RFE-selected features:
- **RandomForestClassifier:** Achieved an accuracy of **0.9768**. Showed strong performance but struggled with some minority classes (e.g., classes 6 and 9).
- **LogisticRegression:** Performed significantly worse, with an accuracy of **0.2830**. This indicates that the relationships in the data are likely non-linear, which Logistic Regression cannot capture effectively.
- **DecisionTreeClassifier:** Achieved an accuracy of **0.9763**, very similar to RandomForest, but also showing weaknesses in minority classes.
- **XGBoostClassifier (Untuned):** Achieved an accuracy of **0.9789**. This was the best performing untuned model, demonstrating strong capabilities for this dataset. Similar to other tree-based models, it also faced challenges with minority classes (6 and 9).

Hyperparameter Tuning (XGBoost)

- Both GridSearchCV and RandomizedSearchCV were used to tune the XGBoost Classifier on the RFE-selected features.
- **GridSearchCV Results:**
 - Best Parameters: {'learning_rate': 0.1, 'max_depth': 7, 'n_estimators': 100}
 - Best Cross-Validation Accuracy: 0.9801
 - Test Set Accuracy: 0.9791
- **RandomizedSearchCV Results:**
 - Best Parameters: {'learning_rate': 0.081, 'max_depth': 8, 'n_estimators': 178}
 - Best Cross-Validation Accuracy: 0.9800
 - Test Set Accuracy: 0.9791
- Both tuning methods yielded models with virtually identical high accuracy on the test set, suggesting that the initial untuned XGBoost was already performing very well or that the search space explored was robust. RandomizedSearchCV offered a more efficient exploration of the parameter space.

FEATURE IMPORTANCE AND EXPLAINABILITY

MODEL INTERPRETABILITY

(SHAP VS. BUILT-IN IMPORTANCE)

- **SHAP (SHapley Additive exPlanations)** was used to interpret the best-performing XGBoost model.
- **Global Feature Importance (SHAP):** Features like Avg_Fwd_Packet_Size, fwd_header_size_tot, flow_iat.max, and flow_duration were highlighted as most impactful by SHAP, accounting for interactions.
- **Built-in Feature Importance (XGBoost):** flow_duration was ranked highest by the model's internal gain metric, followed by id.resp_p.
- **Comparison Insight:** While both methods identified many common important features, their rankings differed. SHAP provided a more comprehensive measure by considering interaction effects, revealing features like Avg_Fwd_Packet_Size as highly influential despite lower built-in scores.

LIMITATIONS AND FUTURE WORK

MODEL LIMITATIONS

- **Significant Performance Disparities:** Performance varies drastically across different proto/service groups, indicating fairness concerns.
- **Persistent Failure for Minority Attack Types:** Even with SMOTE, the model consistently fails to classify extremely rare attack types (e.g., classes 5, 7, 9) both overall and within specific demographic groups.
- **Limited Effectiveness of SMOTE for Extreme Imbalance:** SMOTE alone is insufficient for classes with extremely low support.
- **Trade-off in Feature Inclusion:** Forcing the inclusion of 'proto' and 'service' for fairness analysis slightly reduced overall accuracy and sometimes worsened minority class performance, suggesting that feature selection must balance predictive power with fairness goals.

Future Work:

1. **Enhanced Handling of Class Imbalance:** The consistently low performance on very rare attack categories, despite SMOTE application, suggests the need for more advanced methods.
2. **Domain-Specific Anomaly Detection:** For attack classes with minimal representation (such as classes 5, 7, 9, which repeatedly showed 0.00 metrics), it may be more effective to consider them as anomalies rather than separate classes. Approaches like Isolation Forest, One-Class SVM, or Autoencoders warrant further investigation.
3. **Hierarchical Classification:** Considering the characteristics of attack types, implementing a hierarchical classification strategy could be advantageous. This would involve initially distinguishing between broad groups (e.g., normal vs. attack) and subsequently classifying the 'attack' samples into specific categories. This method could better allocate focus to more challenging attack traffic.
4. **Investigation of Additional Advanced Models:** Although XGBoost has demonstrated strong performance, other robust models might offer benefits:
 - **LightGBM and CatBoost:** These gradient boosting frameworks are recognized for their efficiency and effectiveness, particularly with categorical variables.
 - **Deep Learning Techniques:** For capturing complex network traffic patterns, models such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) could be employed to extract detailed features, especially when treating data as sequential.

THANK YOU

Manuel Terence P. Manuel

PGDAIML – Capstone Project